www.csvtuonline.com

Printed Pages – 6

324414 (24)

BE (4th Semester)

Examination, April-May, 2014

Branch : Electrical

DIGITAL ELECTRONICS & LOGIC DESIGN

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Each question has 4 parts except Q. No. 1. Part (a) is compulsory & it carries 2 marks. Solve any 2 questions from remaining three parts i.e. (b), (c) & (d), which carries 7 marks each.

UNIT–I

Q. 1. (a) Write first 20 numbers of hexadecimal number system. 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) Perform the following operations :

(i) Convert $2A6_{16}$ to decimal.

(ii) Calculate $84_{16} - 2A_{16}$ in hexadecima

(iii) Calculate $DF_{16} + AC_{16}$ in hexadecim

(c) What is a cyclic code ? How can generate Gray code.

UNIT–II

Q. 2. (a) Explain how A + (B.C) = (A + B).(A + C) true in Boolean Algebra, but not in ordina algebra.

(b) Simplify the following equation usi Karnaugh mapping procedure :

$$X = \overline{A}\,\overline{B}\,\overline{C} + A\overline{C}\,\overline{D} + A\overline{B} + ABC\overline{D} + \overline{A}\,\overline{B}C$$

www.csvtuonline.com

www.csvtuonline.com

(c) Find the set of prime implicants from the switching function given below. Find out the minimised function using Quine-McClusky method. 7

$f(A,B,C,D) = \Sigma(0,5,7,8,9,10,11,14,15)$

(d) Simplify using K-map and give answer in pos form : 7

$F(w,x,y,z) = \Sigma(1,3,7,11,15)$

$d(w,x,y,z) = \Sigma(0,2,5)$

UNIT–III

Q. 3. (a) In a combination circuit, can there be more number of outputs than inputs ? Explain your answer with an example ? 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) Design a combination circuit that acce three-bit number and generate an o binary number equal to the square o input number.

(c) Design a full adder circuit with x & y a inputs, z as input carry. Assume S as and C as carry output.

(d) How ROM can be used to impleme combinational circuit. Make a full a using ROM.

UNIT–IV

Q. 4. (a) What is the difference between a sequen and combinational logic.

www.csvtuonline.com

www.csvtuonline.com

(b) Write down the excitation table for RS, JK, D and T flip flops. 7

(c) Design a counter (3-bit) using T flip-flop. 7

(d) Draw & explain working of a 4-bit register with parallel load. 7

UNIT–V

Q. 5. (a) How the size of a memory unit expressed ? 2

(b) Explain working of memory cell of IC-memory with the help of logic diagram. Also show block-diagram. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) Design a circuit which detects a sequence 010. 7

(d) Design a 3-bit parity generator. 7

www.csvtuonline.com

www.csvtuonline.com